
RATE WARS
The Standardization Saga
Chapter 10: The Direct & Indirect Techniques
In a world of unequal populations, raw data is a trap.
One analyst must level the playing field to find the truth...
Based on "Digging into Math: Calculating Rates and Risks"
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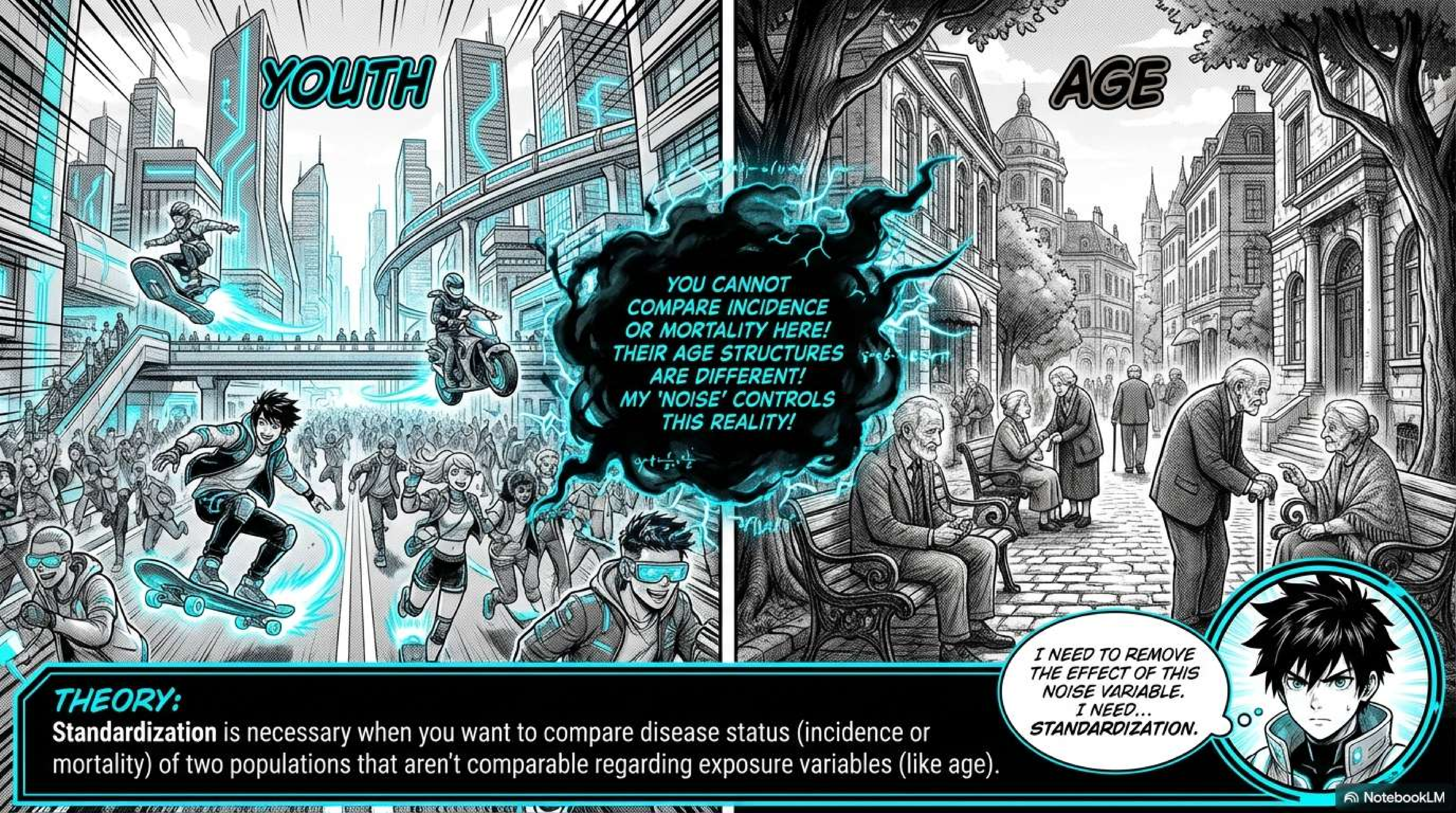
YOUTH
AGE
YOU CANNOT COMPARE INCIDENCE OR MORTALITY HERE! THEIR AGE STRUCTURES ARE DIFFERENT! MY 'NOISE' CONTROLS THIS REALITY!
THEORY:
Standardization is necessary when you want to compare disease status (incidence or mortality) of two populations that aren't comparable regarding exposure variables (like age).
I NEED TO REMOVE THE EFFECT OF THIS NOISE VARIABLE. I NEED... STANDARDIZATION.
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TECHNIQUE 1: THE DIRECT STRIKE
The Direct Method
The Question: What would the mortality rate be if my Study Population had the same age structure as the Standard Population?
CRITICAL INTEL:
To perform this technique, you must select a standard:
1. The least exposed population.
2. A sum of two populations.
3. An international standard.
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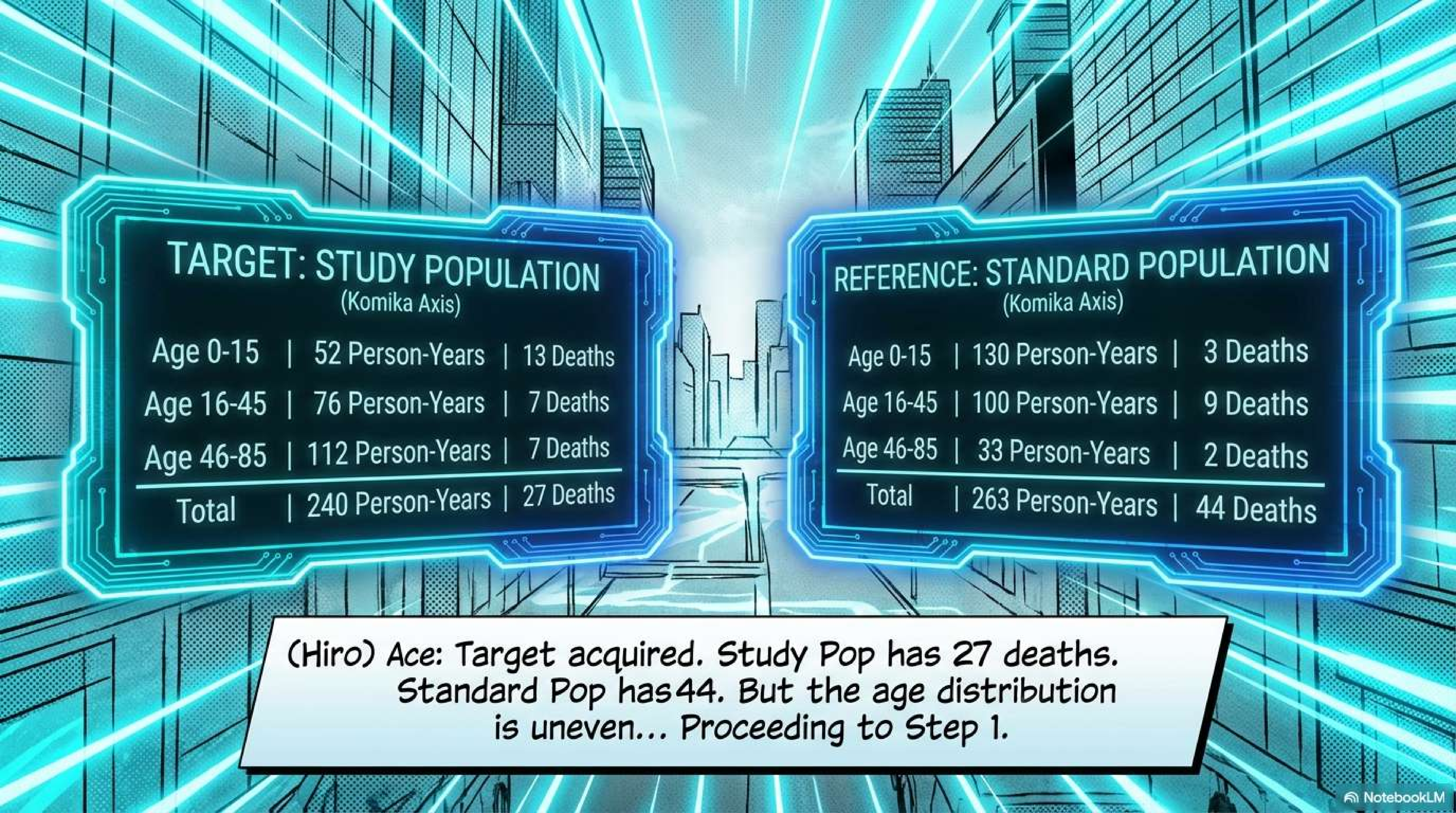

TARGET: STUDY POPULATION
(Komika Axis)
Age 0-15 | 52 Person-Years | 13 Deaths
Age 16-45 | 76 Person-Years | 7 Deaths
Age 46-85 | 112 Person-Years | 7 Deaths
Total | 240 Person-Years | 27 Deaths
REFERENCE: STANDARD POPULATION
(Komika Axis)
Age 0-15 | 130 Person-Years | 3 Deaths
Age 16-45 | 100 Person-Years | 9 Deaths
Age 46-85 | 33 Person-Years | 2 Deaths
Total | 263 Person-Years | 44 Deaths
(Hiro) Ace: Target acquired. Study Pop has 27 deaths. Standard Pop has44. But the age distribution is uneven... Proceeding to Step 1.

STEP 1: THE OBSERVATION

Formula: $O = \dfrac{\text{Total deaths in study}}{\text{Total person years}}$

$$O = 27 / 240$$

$$O = 0.1125$$

First, **calculate the observed death rate** for the study population.
This is the raw power level before adjustment.

STEP 2: WEIGHING THE SOUL

Calculate age-specific weights using the **STANDARD** Population.

Formula: $$\text{Weight} = \frac{\text{Age specific person years (Standard)}}{\text{Total person years (Standard)}}$$

Age 0-15: **130 / 263 = 0.49**

Age 16-45: **100 / 263 = 0.38**

Age 46-85: **33 / 263 = 0.13**

STEP 3: THE AGE-SPECIFIC STRIKE

Calculate age-specific death rates for the STUDY Population.

Age 0-15: 13 / 52 = 0.25

Age 16-45: 7 / 76 = 0.0921

Age 46-85: 7 / 112 = 0.0625

STEP 4: THE EXPECTED RATE (E)

0.25

0.0921

0.0625

0.49

0.38

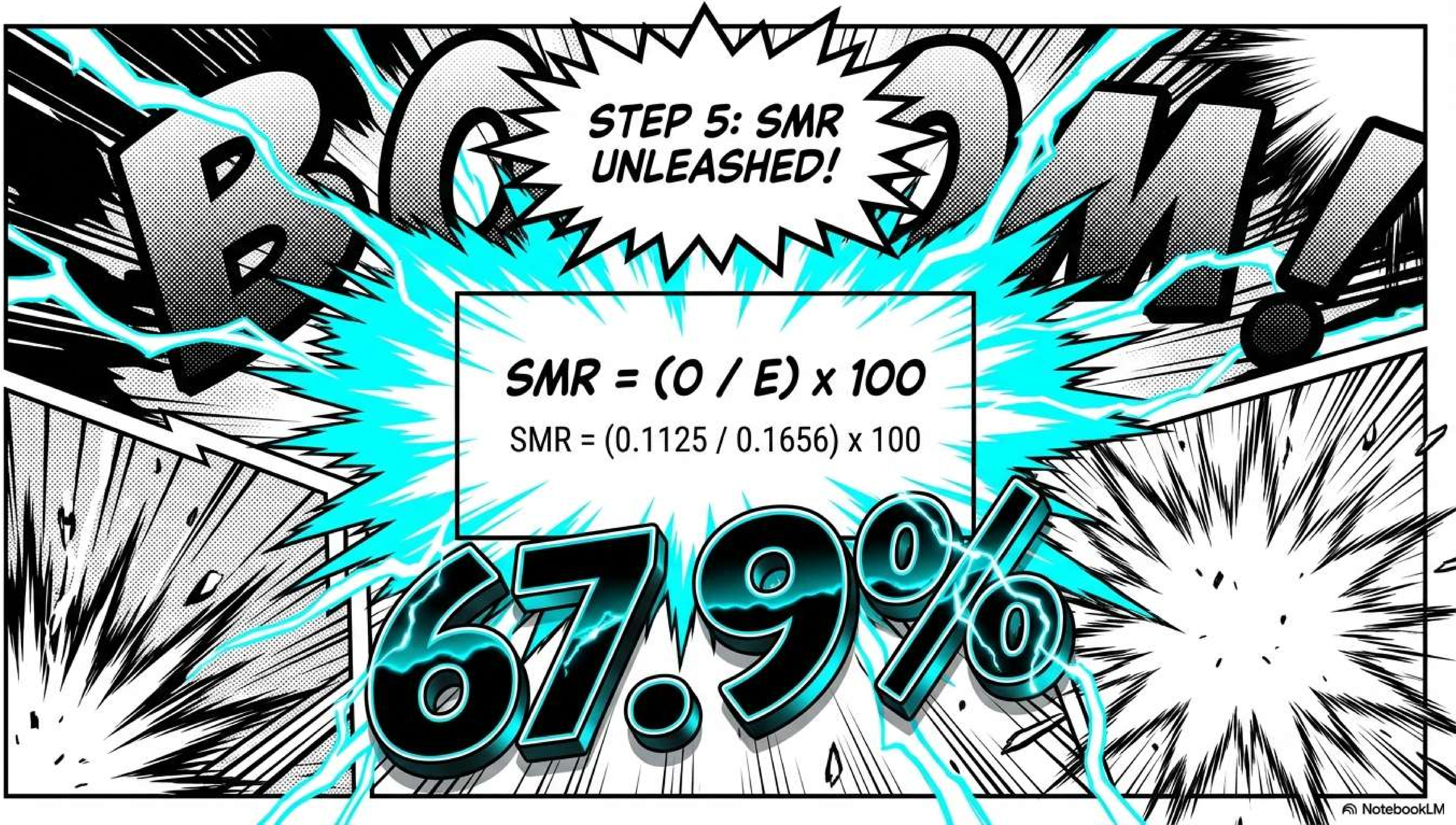
0.13

$$E = \sum \frac{\text{(Age-specific death rate in study population)}}{\text{(Age-specific weight in standard population)}}$$

$$E = (\text{Rate}_{\text{study}} \times \text{Weight}_{\text{standard}}) + \ldots$$

$$E = \underbrace{(0.25 \times 0.49)}_{=\,0.1225} + \underbrace{(0.0921 \times 0.38)}_{=\,0.034998} + \underbrace{(0.0625 \times 0.13)}_{=\,0.008125}$$

$$E = 0.1656$$



STEP 5: SMR UNLEASHED!
SMR = (O / E) x 100
SMR = (0.1125 / 0.1656) x 100
67.9%
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VICTORY ANALYSIS

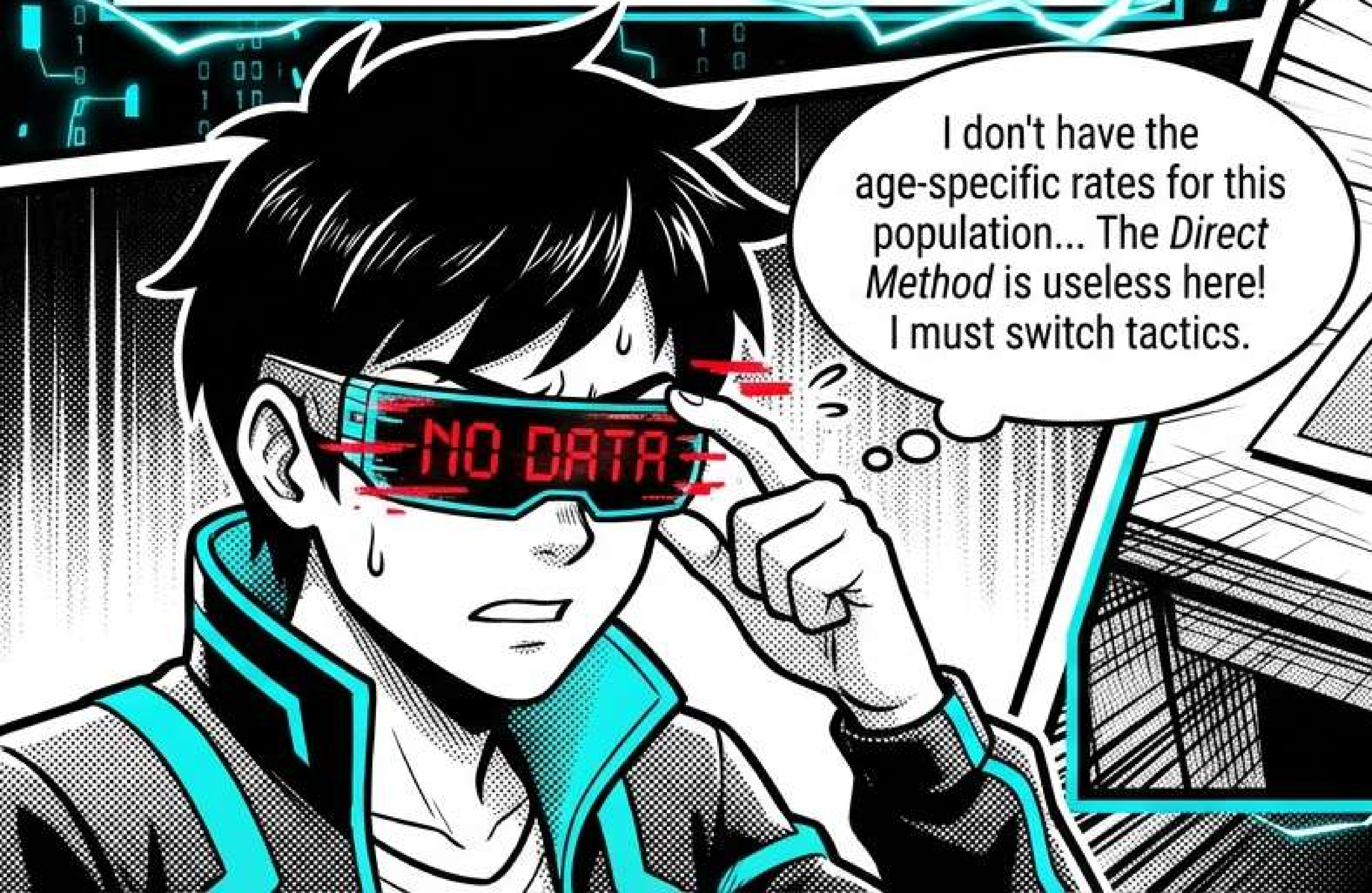
The Result: SMR is 67.9%

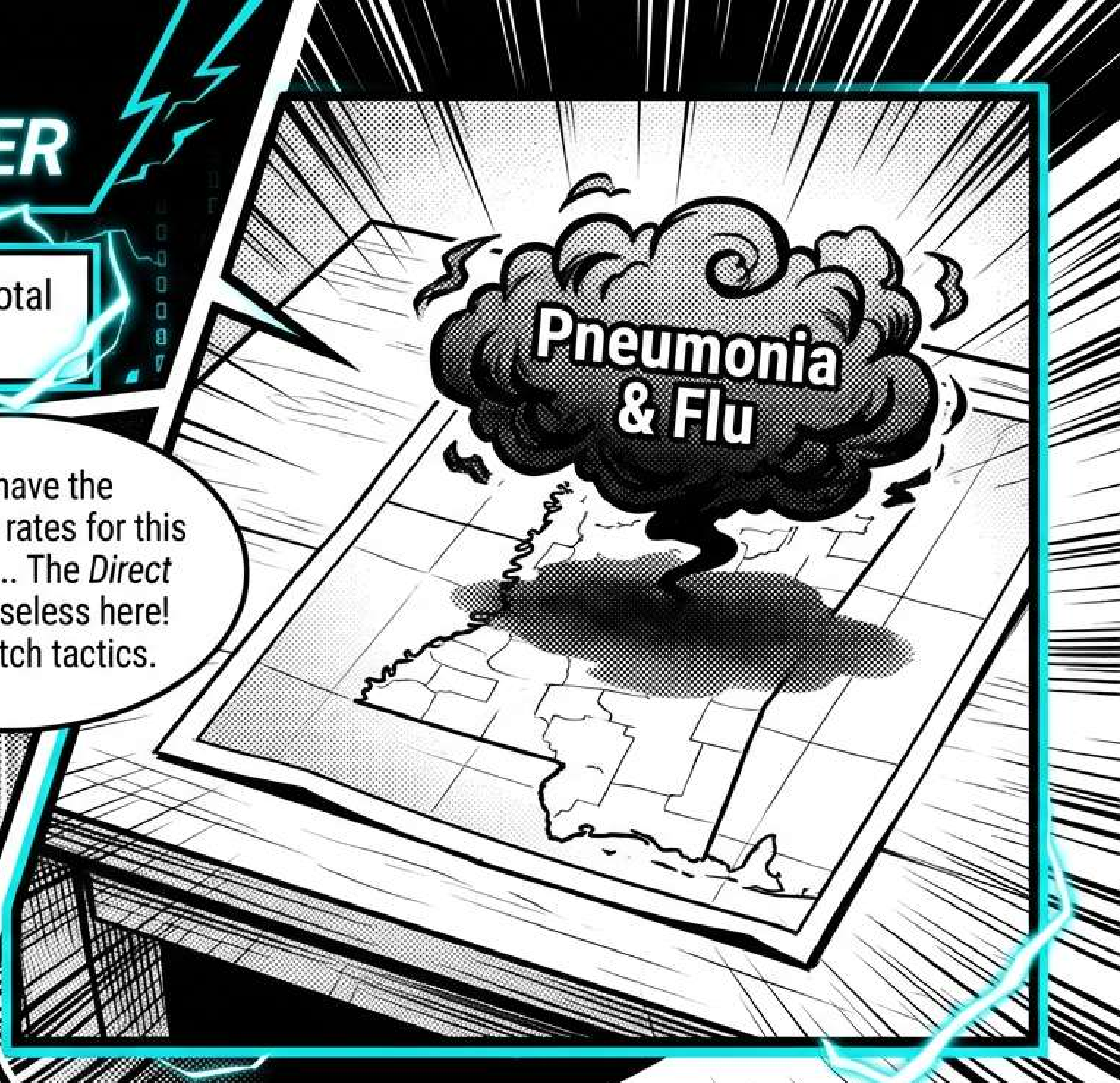
Technical Translation: The expected death rate would be 67.9% higher than the observed death death rate in the study population if the study population had the same age-specific rates as the standard population.

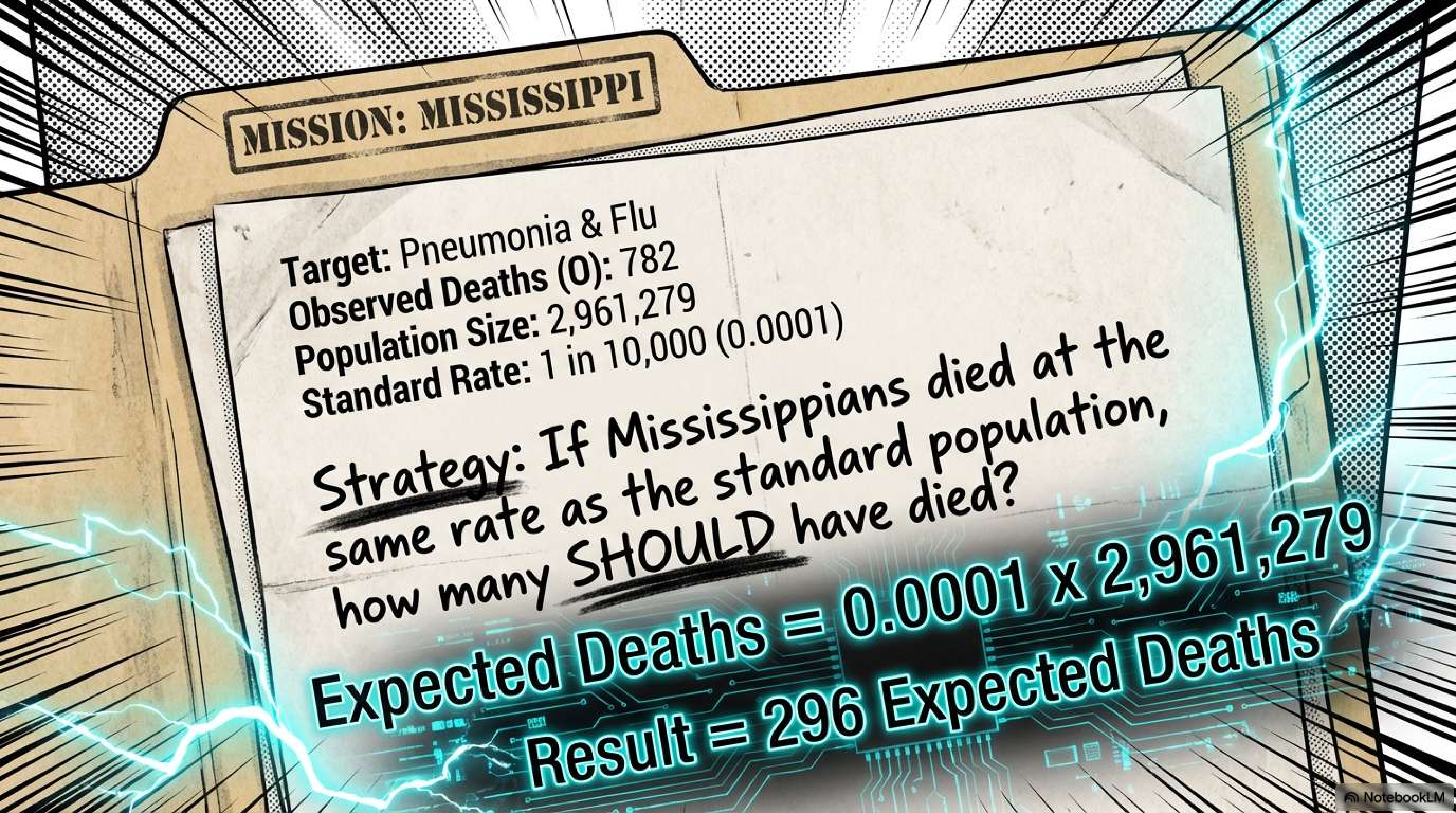
Plain English: Our Observed Rate (O) was lower than the Expected Rate (E). The study population is actually quite healthy compared to the standard!

TECHNIQUE 2: THE INDIRECT COUNTER

Use the **INDIRECT METHOD** when you only have total numbers and the rate of a standard population.






MISSION: MISSISSIPPI
Target: Pneumonia & Flu
Observed Deaths (O): 782
Population Size: 2,961,279
Standard Rate: 1 in 10,000 (0.0001)
Strategy: If Mississippians died at the same rate as the standard population, how many SHOULD have died?
Expected Deaths = 0.0001 x 2,961,279
Result = 296 Expected Deaths
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INDIRECT SMR STRIKE

SMR = Observed Deaths / Expected Deaths

SMR = 782 / 296

SMR = 2.64

SMR = 264%

Interpretation: Observed is 264% of expected! If you take the rate of the standard population, the death rate is expected to be 264 percent lower than the observed rate.

CHOOSE YOUR WEAPON

DIRECT METHOD (Preferred)

- **When to use:** You have age-specific rates for your study population.
- **Mechanism:** Apply Study Rates to Standard Weights.

INDIRECT METHOD (Easier)

- **When to use:** You don't have age-specific rates, just total numbers.
- **Mechanism:** Apply Standard Rates to Study Population count.

KEY INTEL / CREDITS

STANDARDIZATION: Removing the effect of a confounder (noise) like age.

CONFOUNDER: A variable that distorts the relationship between exposure and disease.

SMR: Standardized Mortality Ratio (O / E).